



TECHMINDS ACADEMY

React.js Mastery

Modern Frontend Development Curriculum · Intermediate to Job-Ready

A structured 14-week programme for students who have completed the TechMinds HTML, CSS, and JavaScript curricula. From your first component to deploying a full single-page application on Vercel.

14 Weeks	5 Stages	65+ Lessons	9 Projects	3 Prerequisites
--------------------	--------------------	-----------------------	----------------------	---------------------------

JSX & Components · Hooks · React Router · Zustand · React Hook Form · useMemo · Testing · Vercel Deployment

LEARNING PATHWAY

Programme Overview

Stage	Focus	Duration	Level	Prerequisite
Stage 1	React Foundations — JSX, components, state	3 weeks	Beginner	HTML+CSS +JS
Stage 2	Core Hooks & Component Patterns	3 weeks	Beginner	Stage 1
Stage 3	React Router & Navigation	2 weeks	Intermediate	Stage 2
Stage 4	State Management & Forms	3 weeks	Intermediate	Stage 3
Stage 5	Performance, Testing & Capstone	3 weeks	Advanced	Stage 4

CURRICULUM RATIONALE

Key Decisions Explained

Why React?	React is the most widely used frontend library in the world. It is the first thing most companies look for in a frontend developer. Mastering React opens doors to React Native (mobile), Next.js (full-stack), and most modern frontend roles.
Why Zustand over Redux?	Redux has a steep learning curve and significant boilerplate. Zustand achieves the same result — predictable global state — with far less code. Students learn the concept correctly and can pick up Redux easily later if needed.
Why include testing?	Testing is a skill most bootcamps skip and most employers value. Introducing React Testing Library in Week 13 ensures students understand how to verify their code works — a quality that stands out in job applications.
What comes after this course?	React Mastery feeds directly into the TechMinds Next.js course — covering server-side rendering, the App Router, and full-stack development. Students can also move to the React Native track for mobile development.

Stage 01 · Beginner

3 Weeks · Weeks 1 – 3

React Foundations

Thinking in components and writing your first React app

This stage assumes the student has completed the TechMinds HTML, CSS, and JavaScript curricula. React is introduced from scratch — what it is, why it exists, and how to think in components. By the end of Week 3 the student will have a working React app running in the browser with interactive state.

Week 1 Introduction to React & JSX

5 lessons · 6 hrs

[01](#) What is React? Why companies use it[02](#) How React differs from vanilla JavaScript[03](#) Setting up a React project with Vite[04](#) JSX — writing HTML inside JavaScript[05](#) JSX rules: className, self-closing tags, expressions in {}[06](#) Your first React component — function that returns JSX**PROJECT**

A personal introduction card component — name, role, bio, and avatar

Week 2 Components & Props

5 lessons · 6 hrs

[01](#) What are components? Thinking in building blocks[02](#) Creating and rendering multiple components[03](#) Props — passing data from parent to child[04](#) Destructuring props for cleaner code[05](#) Default props — fallback values when none are passed[06](#) PropTypes — basic type checking for props**PROJECT**

A product card component — reused 4 times with different props for each product

Week 3 State & Event Handling

5 lessons · 7 hrs

[01](#) What is state? Why React needs to track it[02](#) useState hook — declaring and updating state[03](#) How React re-renders when state changes[04](#) Handling onClick and onChange events[05](#) Controlled inputs — syncing form fields with state[06](#) Lifting state up — sharing state between components**PROJECT**

An interactive shopping cart — add items, update quantities, show running total

Stage 02 · Beginner

3 Weeks · Weeks 4 – 6

Core Hooks & Component Patterns

Mastering React hooks and building reusable component patterns

Hooks are the heart of modern React. This stage covers the most essential hooks in depth — `useEffect` for side effects, `useRef` for DOM access, and `useContext` for sharing data without prop drilling. Students also learn how to render lists and handle conditional UI — skills used in every React application.

Week 4 **useEffect & Side Effects**

5 lessons · 7 hrs

01 What are side effects in React?

02 `useEffect` hook — running code after render

03 The dependency array — controlling when effects run

04 Cleanup functions — avoiding memory leaks

05 Fetching data with `useEffect` and displaying it

06 Loading and error states for async data

PROJECTA live GitHub profile viewer — fetch and display a user's repos using `useEffect`

Week 5 **Lists, Keys & Conditional Rendering**

5 lessons · 6 hrs

01 Rendering lists with `array.map()` in JSX02 The `key` prop — why React requires it and how to use it03 Conditional rendering with `&&` and the ternary operator

04 Rendering nothing — returning null from a component

05 Dynamic class names based on state or props

06 Building a filterable list — show and hide items

PROJECT

A filterable task board — tasks displayed as cards, filter by status (All / Active / Done)

Week 6 **useRef & useContext**

5 lessons · 6 hrs

01 `useRef` — accessing DOM elements directly in React02 `useRef` for persisting values without triggering re-renders03 Practical `useRef`: focus an input, scroll to a section

04 The prop drilling problem — why it gets painful

05 `useContext` — creating and consuming a React context

06 Building a theme switcher (light/dark) with context

PROJECT

A dark mode app — toggle light and dark theme globally using Context across multiple components

Stage 03 · Intermediate

2 Weeks · Weeks 7 – 8

React Router & Navigation

Building multi-page React applications with client-side routing

Single-page applications need navigation without full page reloads. This stage introduces React Router — the standard routing library for React. Students learn to define routes, navigate between pages, pass data via URL parameters, and protect routes that require authentication.

Week 7

React Router Basics

5 lessons · 7 hrs

01 What is client-side routing? How it differs from server routing

02 Installing and setting up React Router v6

03 BrowserRouter, Routes, and Route components

04 The Link and NavLink components — navigation without page reload

05 URL parameters with useParams — dynamic routes

06 The useNavigate hook — programmatic navigation

PROJECT

A 4-page app: Home, About, Blog list, and Blog detail — all linked with React Router

Week 8

Advanced Routing Patterns

5 lessons · 7 hrs

01 Nested routes — layouts shared across pages

02 The Outlet component — rendering child routes

03 Index routes — default content for a parent route

04 Query parameters with useSearchParams — filtering via URL

05 Protected routes — redirecting unauthenticated users

06 404 pages — handling unknown routes gracefully

PROJECT

A protected dashboard — login page, redirect on auth, nested route layout with sidebar

Stage 04 · Intermediate

3 Weeks · Weeks 9 – 11

State Management & Forms

Managing complex state and building production-grade forms

As applications grow, state becomes complex. This stage teaches `useReducer` for predictable state updates and introduces `Zustand` — a lightweight global state manager used widely in production. Forms are covered in depth, including validation, error messages, and the `React Hook Form` library.

Week 9 useReducer & Complex State

5 lessons · 7 hrs

- 01 When `useState` is not enough — the case for `useReducer`
- 02 `useReducer` — state, action, and dispatch
- 03 Writing a reducer function — handling multiple action types
- 04 Combining `useReducer` with `useContext` for global state
- 05 `Immer` — writing immutable updates in a mutable style
- 06 Comparing `useState`, `useReducer`, and external state libraries

PROJECT A shopping cart with `useReducer` — add, remove, increment, decrement, and clear cart

Week 10 Global State with Zustand

5 lessons · 7 hrs

- 01 What is global state management and when do you need it?
- 02 Installing and setting up `Zustand`
- 03 Creating a store — state and actions in one place
- 04 Consuming the store in any component without prop drilling
- 05 Persisting state to `localStorage` with `Zustand` middleware
- 06 `Zustand` vs `Context` — when to use each

PROJECT A notes app with `Zustand` — create, edit, delete, and pin notes; state persists on refresh

Week 11 Forms & Validation

5 lessons · 6 hrs

- 01 Controlled forms with `useState` — the manual approach
- 02 The problems with manual form handling at scale
- 03 `React Hook Form` — fast, flexible, and uncontrolled
- 04 `register`, `handleSubmit`, and `formState.errors`
- 05 Validation rules: `required`, `minLength`, `pattern`, `validate`
- 06 Integrating `React Hook Form` with a UI component library

PROJECT A multi-step registration form — personal info, account setup, confirmation; fully validated

Stage 05 · Advanced

3 Weeks · Weeks 12 – 14

Performance, Testing & Capstone

Optimising React apps and shipping a production-quality project

The final stage covers the skills that separate junior developers from mid-level ones — performance optimisation with `useMemo` and `useCallback`, custom hooks for reusable logic, and writing tests with React Testing Library. The course closes with a capstone project: a full, deployed React application.

Week 12 Performance & Custom Hooks

5 lessons · 7 hrs

- 01 React rendering — when and why components re-render
- 02 `useMemo` — caching expensive computed values
- 03 `useCallback` — caching functions to prevent child re-renders
- 04 `React.memo` — skipping re-renders for unchanged components
- 05 Custom hooks — extracting and reusing stateful logic
- 06 Building `useFetch`, `useLocalStorage`, and `useDebounce` hooks

PROJECT Refactor the Week 10 notes app using custom hooks and memoisation

Week 13 Testing React Components

5 lessons · 7 hrs

- 01 Why testing matters — catching bugs before users do
- 02 Setting up Vitest and React Testing Library
- 03 Rendering components in tests with `render()`
- 04 Querying the DOM: `getByText`, `getByRole`, `getByPlaceholderText`
- 05 Simulating user interactions with `userEvent`
- 06 Testing async components — `waitFor` and `findBy` queries

PROJECT Write a full test suite for the Week 3 shopping cart — `render`, `interact`, `assert`

Week 14 Capstone Project

6 sessions · 12 hrs

- 01 Planning the app — features, routes, state, and API
- 02 Project setup — Vite, React Router, Zustand, folder structure
- 03 Building the UI — components, layout, and responsive styling
- 04 Connecting the API — `fetch`, loading states, error handling
- 05 Adding auth — login, protected routes, and persisted session
- 06 Deployment — build for production and deploy to Vercel

PROJECT A full React SPA — student chooses: job board, e-commerce store, or social bookmarking app; deployed to Vercel

ASSESSMENT & COMPLETION

How Students Are Evaluated

Component	Description	Weight
Weekly Projects (×9)	React component or feature built from scratch each week	40%
Stage Assessments (×5)	Short quiz and code review at end of each stage	30%
Capstone Application	Full React SPA — built, tested, and deployed to Vercel	30%

LEARNING OUTCOMES

What Students Can Do on Completion

On completing this programme, the student will be able to:

- Build React applications from scratch using Vite and modern tooling
- Design and compose reusable components with props and state
- Use React hooks: useState, useEffect, useRef, useContext, useReducer, useMemo, useCallback
- Write custom hooks to encapsulate and reuse stateful logic
- Build multi-page SPAs with React Router v6 including protected routes
- Manage global application state cleanly with Zustand
- Build validated, multi-step forms using React Hook Form
- Write component tests using Vitest and React Testing Library
- Deploy a production React application to Vercel

TOOLS & RESOURCES

Everything the Student Needs

Tool	Purpose	Cost
VS Code	Primary code editor with React and JSX extensions	Free
Vite	Fast React project scaffolding and build tool	Free
React Router v6	Client-side routing for single-page applications	Free
Zustand	Lightweight global state management library	Free
React Hook Form	Performant, flexible form handling and validation	Free
Vitest + RTL	Unit and component testing for React applications	Free

Vercel	Deploy and host the capstone React app publicly	Free tier
Chrome DevTools	React DevTools extension for inspecting component trees	Free

© 2026 TechMinds Academy. All Rights Reserved. This curriculum is the intellectual property of TechMinds Academy, Bwari, Abuja, Nigeria.
Reproduction or distribution without written permission is prohibited.